Meta’s Custom AI Silicon: From Recommendation to Dual-Mandate with GenAI

MTIA 300 & MTIA 400

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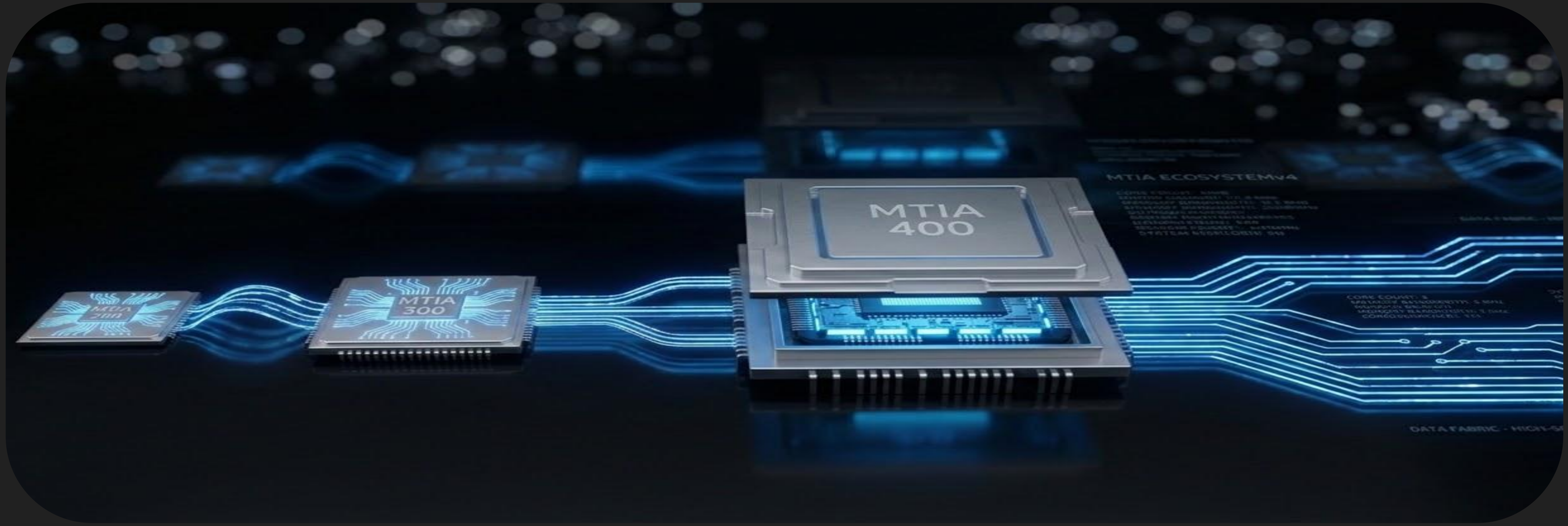


MTIA Program Roadmap

MTIA 200 (2024) [ISCA'25]

Inference accelerator

5nm | 354 TOPS

MTIA 300 (2025) [ISCA'26]

First DLRM training chip

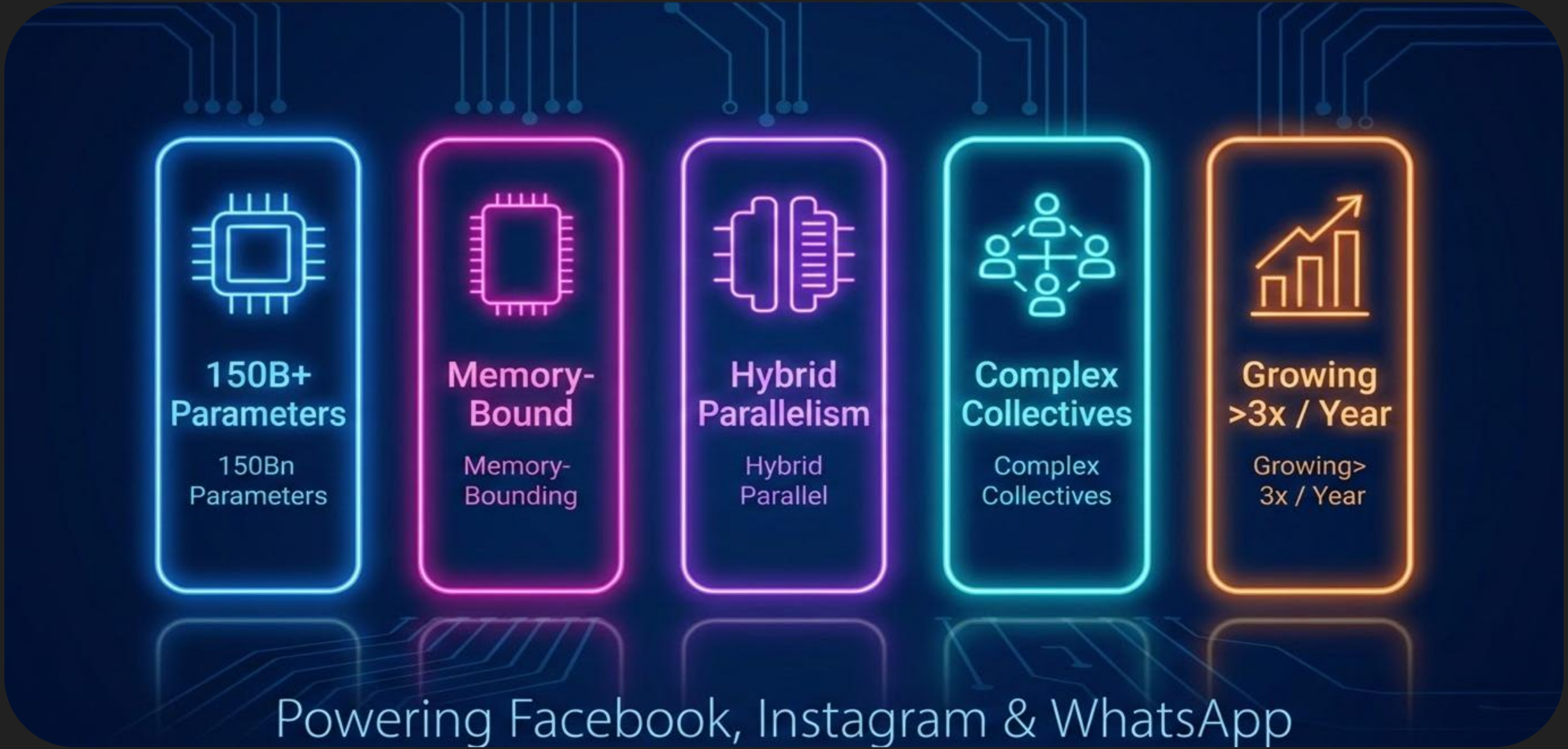
3nm + HBM3E | 1.12 PFLOPS

MTIA 400 (2026+)

DLRM Inference & GenAI training

Deep Dive today

The Meta DLRM Training Challenge



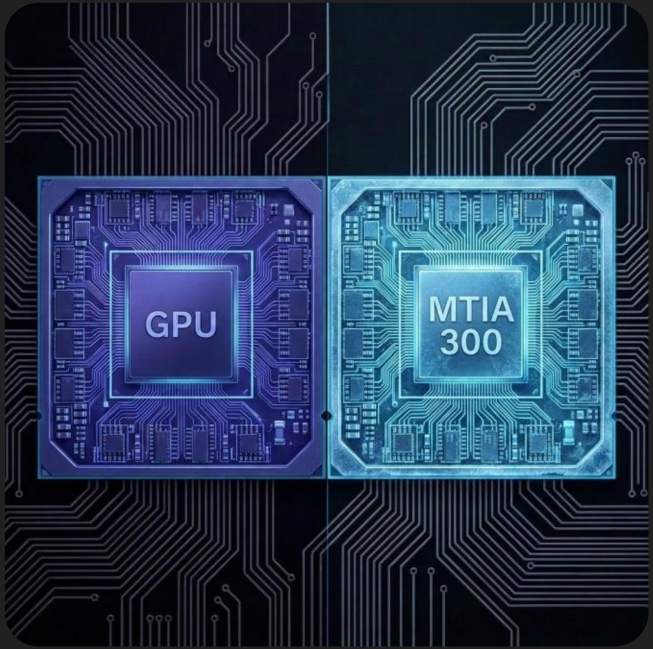
Designed for DLRM Training

The GPU gap for DLRM training:

- Memory-bound sparse ops starve compute
- Low Model FLOPS Utilization (MFU)
- Communication overhead stalls training
- Cost prohibitive at Meta's scale

The MTIA approach:

- Sparse embedding acceleration
- Higher bytes-to-FLOPS ratio (>2x GPU)
- Dedicated collective communication HW
- HW/SW co-design with PyTorch
- Optimized for TCO, not just peak FLOPS



MTIA 300: Meta's First Training Chip

- **1.12 PFLOPS** (FP8) peak compute
- **216 GB HBM3E** at 6.1 TB/s bandwidth
- **1.2 TB/s I/O** via custom on-package RDMA
- **72 Processing Elements (PE)** in 12x6 grid
- **16 Messaging Elements (ME)** for decoupled collectives
- **667W TDP** — liquid-cooled
- 3nm compute + 5nm I/O chiplets + 6x HBM3E

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MTIA 300 Performance

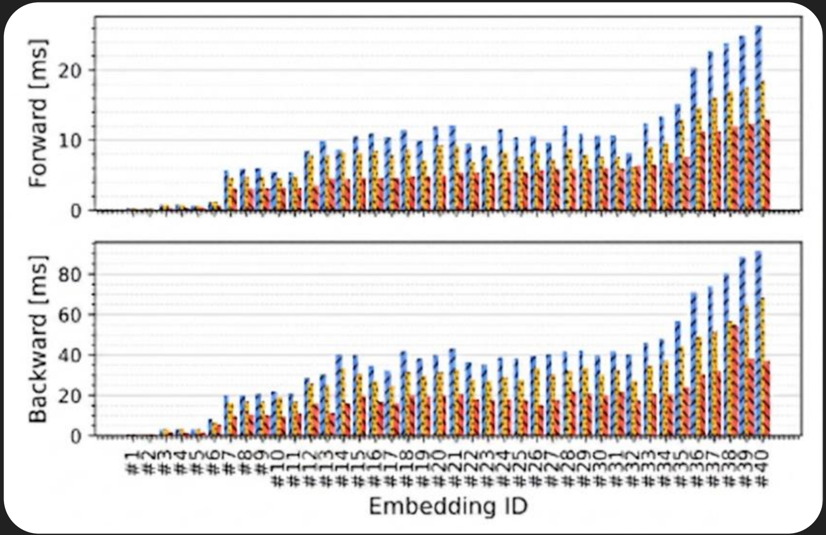
GPU parity on 150B-parameter DLRM model at competitive TCO

Key results:

- 1.87x / 1.88x speedup on embedding operators (fwd/bwd) over leading GPUs
- 1.5x point-to-point bandwidth at 1MB
- 2.2x higher scale-up bandwidth vs GPUs
- 9% Perf/TCO gain from large batch training

Enabled by co-design:

- Large batch (8192 vs 6144) with 216 GB HBM
- Full-precision comms (no FP8 quantization needed)



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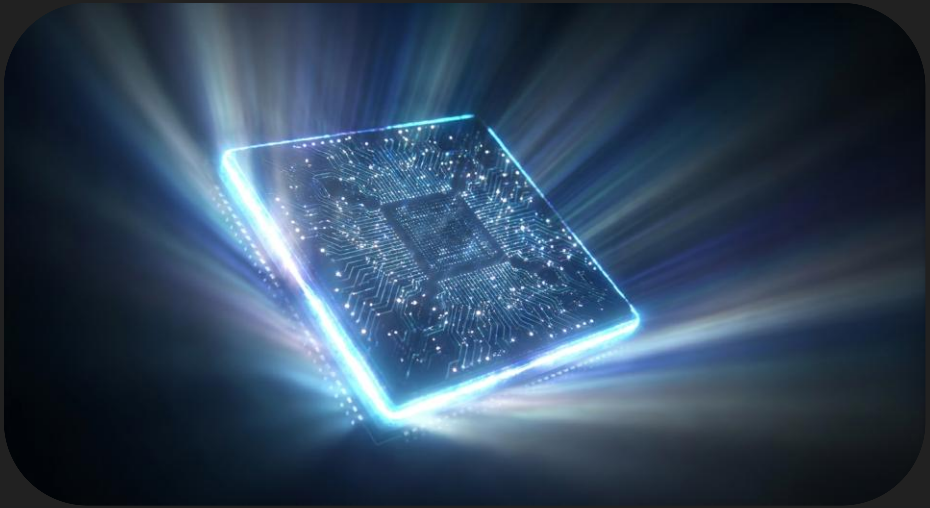
Introducing MTIA 400

Architecture for GenAI + Recommendation Systems



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Motivation for MTIA 400

assignment

Dual Mandate

Support for diverse workloads on a single hardware platform.

- Deep Learning Rec Model (DLRM)
- GenAI Training

memory

MTIA 300 – Optimized for DLRM

- Large sparse embedding tables
- Memory-bound operations

speed

GenAI/LLM Demands

New requirements for advanced AI model scaling.

- Dense compute throughput
- Multi-thousand accelerator scaling

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What's New/Improved in MTIA 400

Multi-chiplet design

- 2 Compute + 1 SoC + 2 I/O
- Higher yield design
- Scale-up cost-effectively

Enhanced compute

- 1.3x num of PEs
- 4x MAC boost for GEMM
- Native MX microscaling
- Horizontal reduction

Special HW for collectives

- Scale-in DMA & streaming reductions
- HW collective synchronization

Memory Hierarchy

- Embedding Caches for DLRM
- In-Memory Reduction in LLC

NoC & RoCE I/O

- Virtual channels for QoS
- 1.2 TBps Scale-up
- 100 GB/s scale-out

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System & Package

5-chiplet 2.5D package

- 2 Compute (3nm, 1.7 GHz): PEs, MEs, on-chip memory
- SoC (3nm, 1.5 GHz): host bridge + control processors
- 2 I/O: scale-up/scale-out networking on RoCE

8 HBM3E stacks

- 288 GB
- 9200 MHz
- 9.4 TB/s peak BW

Die-to-die interconnect

- Compute ↔ Compute: 1.3 TB/s
- Compute ↔ SoC or I/O: 1.2 TB/s

Host interface (SoC chiplet)

- 4×16-lane PCIe Gen6
- 512 GB/s total

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Compute Architecture

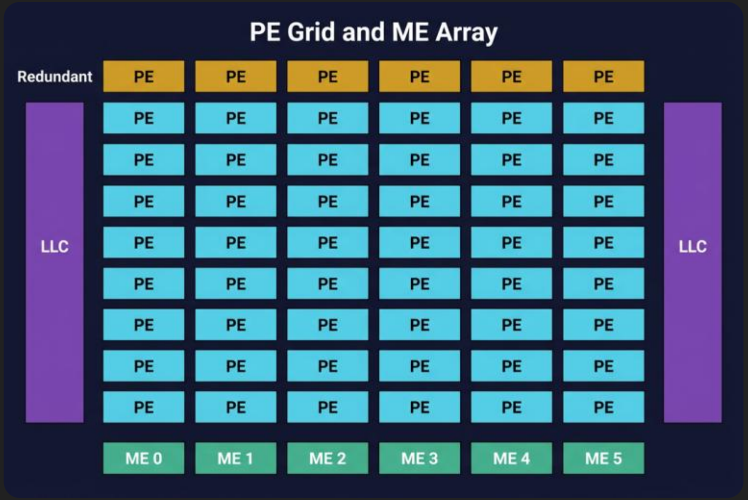
Grid & Array Configuration

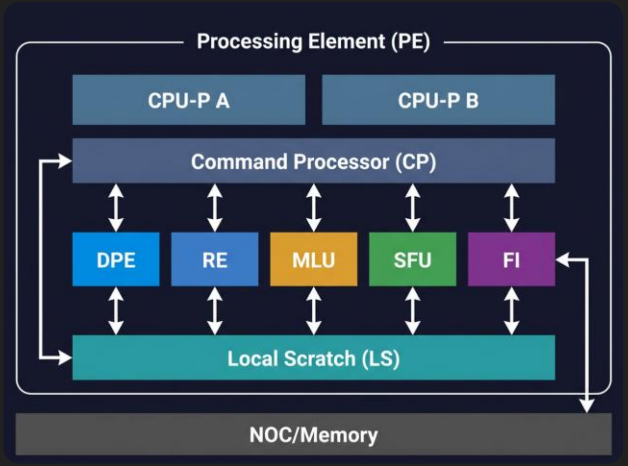
PE grid for ML compute: 8x6 active PEs/chiplet, plus one redundant row

ME array for collectives: 1x6 MEs on south edge

Processing Element (PE) Sub-systems

CPU-P A/B: RISC-V Scalar/Vector
CP: CPU cmd forwarding, DMA
LS: local scratch
DPE: Dot-Product to Compute Registers(CREG)
RE: cross-PE reduction, output CREG to LS
MLU: transpose, LS-to-LS copy
SFU: NL functions, tensor/vector ops, radix sort
FI: fabric interface communicates with the NoC





PE Compute — DPE/RE

DPE/RE Combo – GEMM Engine

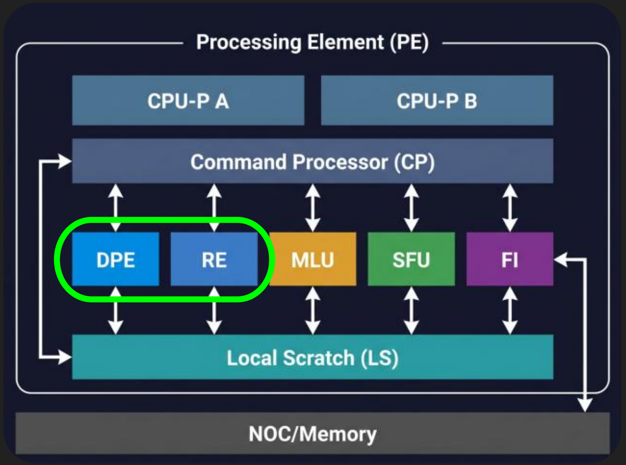
- Together they perform GEMM and output to LS
- Accumulation in compute registers (CREGs)

Leap in GEMM Perf from MTIA 300

- 4x compute HW density
- Entire 256x256 work unit buffered in CREGs

GEMM Input Formats

- BF16, FP16, FP8, FP32(TF32),
- MX8, MX8S, MX4



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PE Compute — SFU

Non-GEMM Tensor/Vector Compute

- Elementwise add/mul/compare
- Non-linear functions: exp, inv, sigmoid, tanh, etc.
- Data type conversions & special functions (acc, min/max)

HW Horizontal Reduction

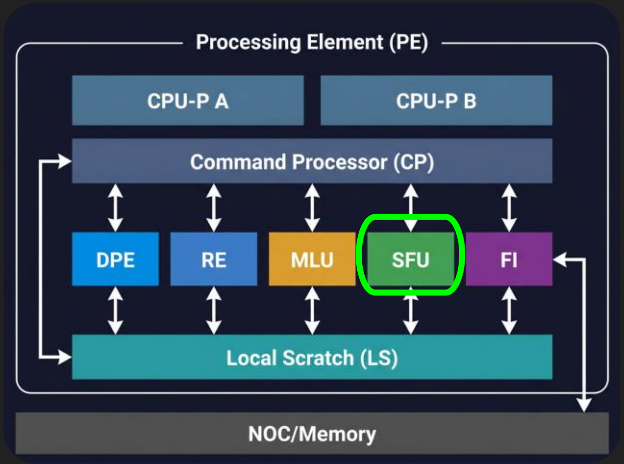
- Optimized for DLRM/GenAI (Layer Norm, SoftMax, RMS Norm)
- MTIA 200/300: Vertical reduction only (requires transpose)
- **MTIA 400:** Added horizontal acc and min/max support

Dedicated Gather Op

64 elements/cycle (HSTU in RMS)

MX Support

MX4/MX8 packing and unpacking



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Microscaling (MX) Formats Support

Native HW Support (OCP MX)

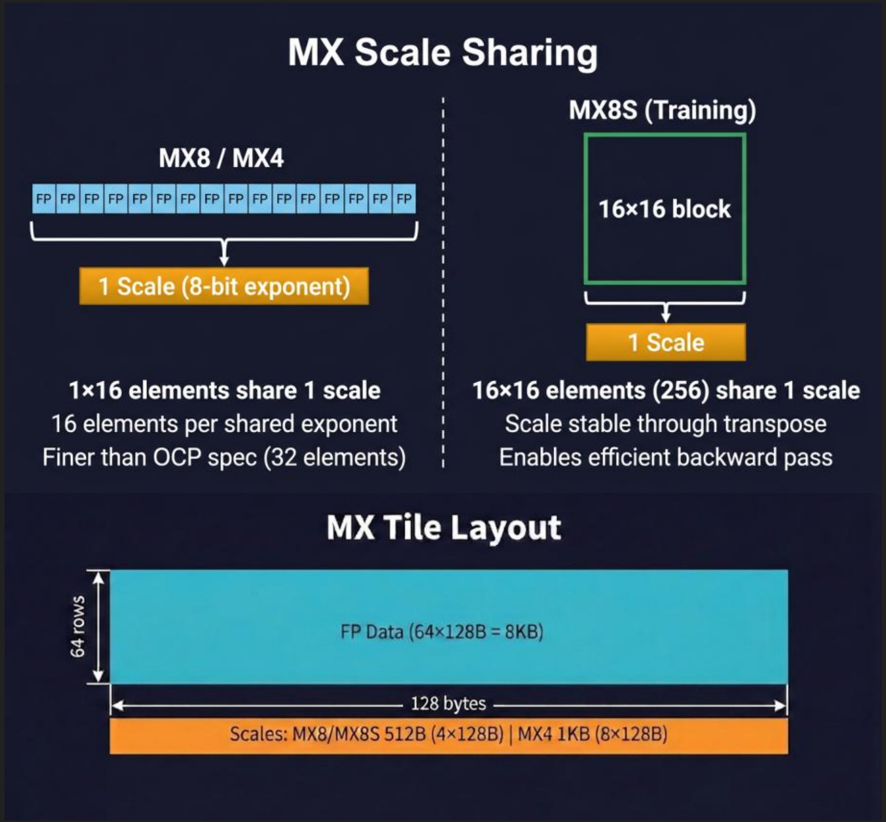
- Finer granularity than spec
- 16 elements per shared exponent (vs. OCP 32)
- SFU converts, DPE consumes

Supported Formats

- MX4 — 4-bit FP, 2× FP8 throughput
- MX8 — 8-bit FP, better dynamic range than raw FP8
- MX8S — 16×16 FP8 block for training

Tiled Layout

- 64×128B FP data tile + scales
  - MX8 512B, MX4 1KB
  - MX8S: fewer scales, same footprint



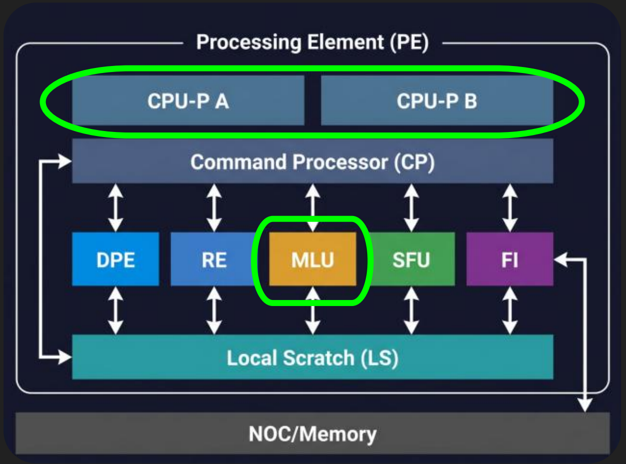
PE Compute — Other Units

2× CPU-P

- Each contains a RISC-V scalar and a vector core
- Custom instruction issue
- General purpose vector operations
- Address bus expanded for HEC

Memory Layout Unit (MLU)

- LS to LS copy
- LS layout transforms: transpose, reshape, concat...



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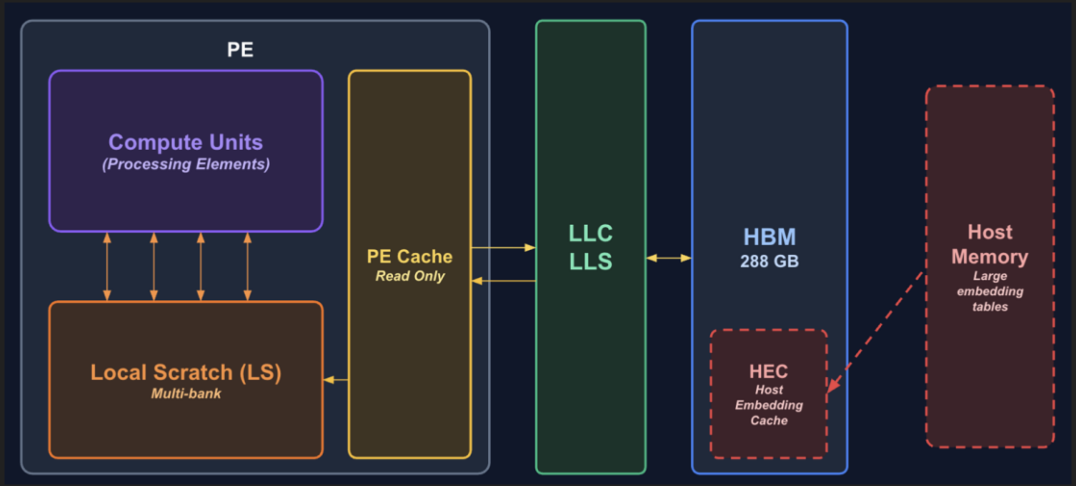
Memory Hierarchy

Multi-tier hierarchy

- **Off-chip memory** : (8 stacks HBM3E) provides 9.4 TB/s BW
- **On-chip SRAM (LLC)** : Configurable partitions/ways; in memory reduction
- **Per-PE Local Scratch** : SW managed; split into Circular Buffers;

Host Embedding Cache (HEC):

- Read only cache for large embedding tables.
- Paired with **PEC** to cache hot embeddings close to PE.



Network-on-Chip (NoC)

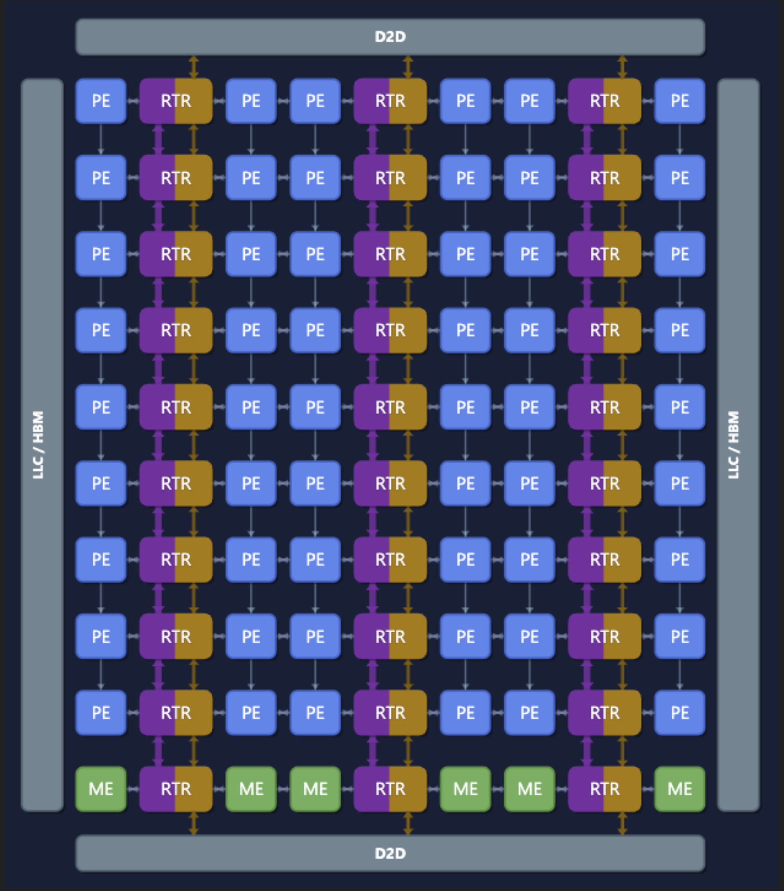
2D Mesh Network

Support multiple VCs to avoid contention among various traffic classes.

- Traffic:
  - Compute : PE ↔ LLC ↔ HBM paths
  - IO : Host PCIe, D2D, and network chiplet
- Auxiliary:
  - Reduction network; Synchronization
  - Job dispatch; debug messaging

NoC Congestion Control:

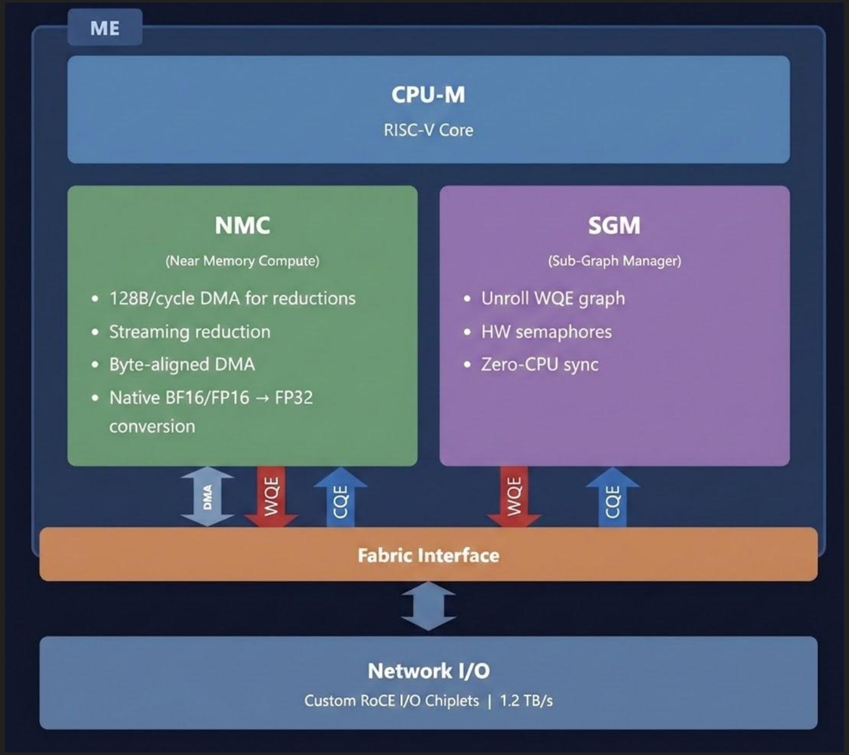
- Leaky Buckets
- Max OT



Enhanced ME – Collective Engine

Dedicated ME array offloads collectives from compute PEs

- 12 MEs/device (6/compute chiplet); RISC-V
- **NMC**: 128 B/cycle DMA; Streaming reduction
- **SGM**: Unroll WQE graph; HW semaphores; Zero-CPU sync
- Offload the Synchronization to HW using
  - Work Queue Entry(WQE)
  - Completion Queue Entry(CQE)



Collective Flow

SEND/RCV Flow

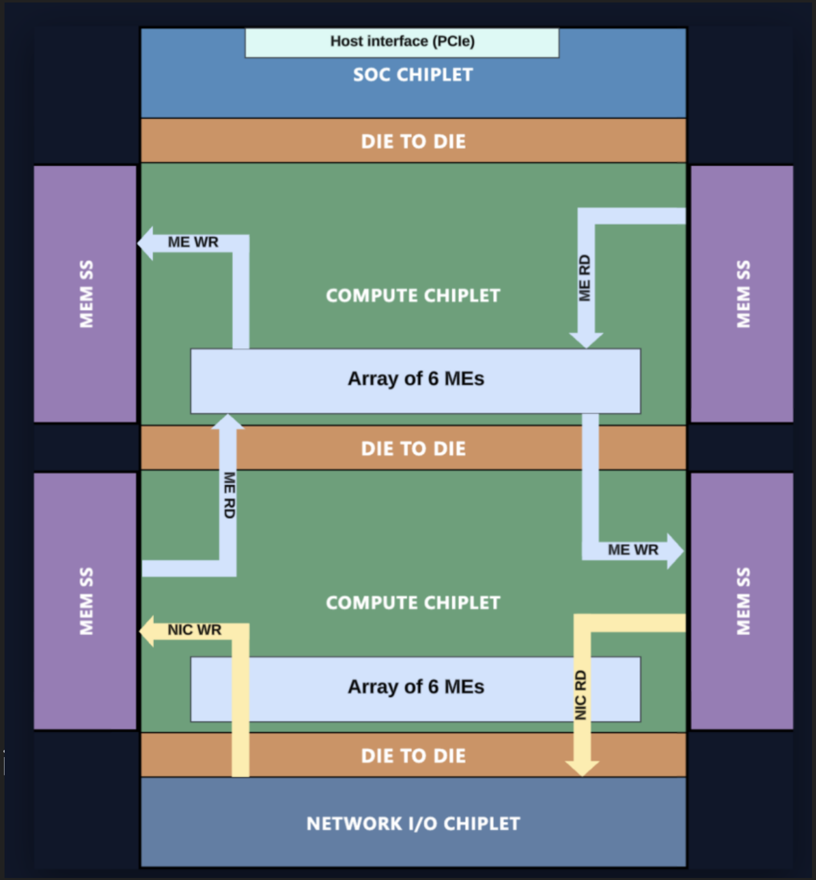
- **Scale-up:** RDMA over NICs @ 1.2TBps
- **Scale-out:** Through PCIe switch

Reduce Flow

- Data reduced in ME or LLC.
- AllReduce/AllToAll for GenAI training

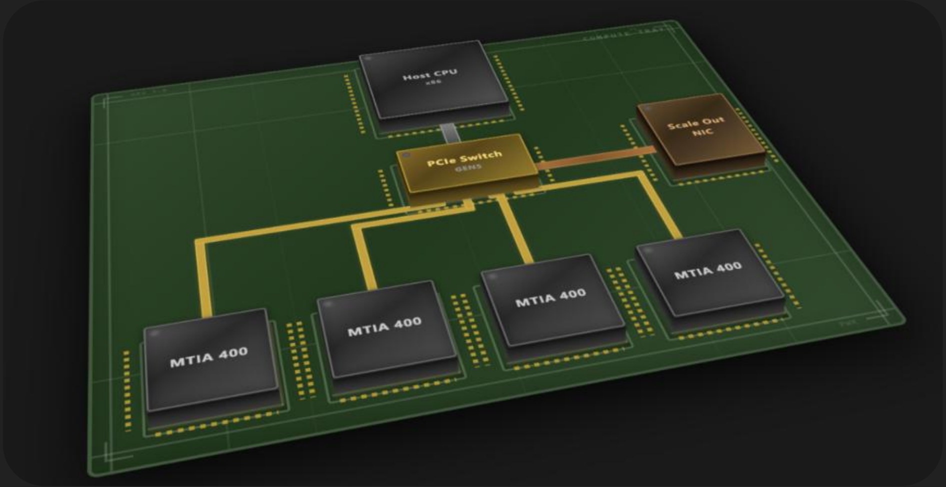
In-Memory Reduction

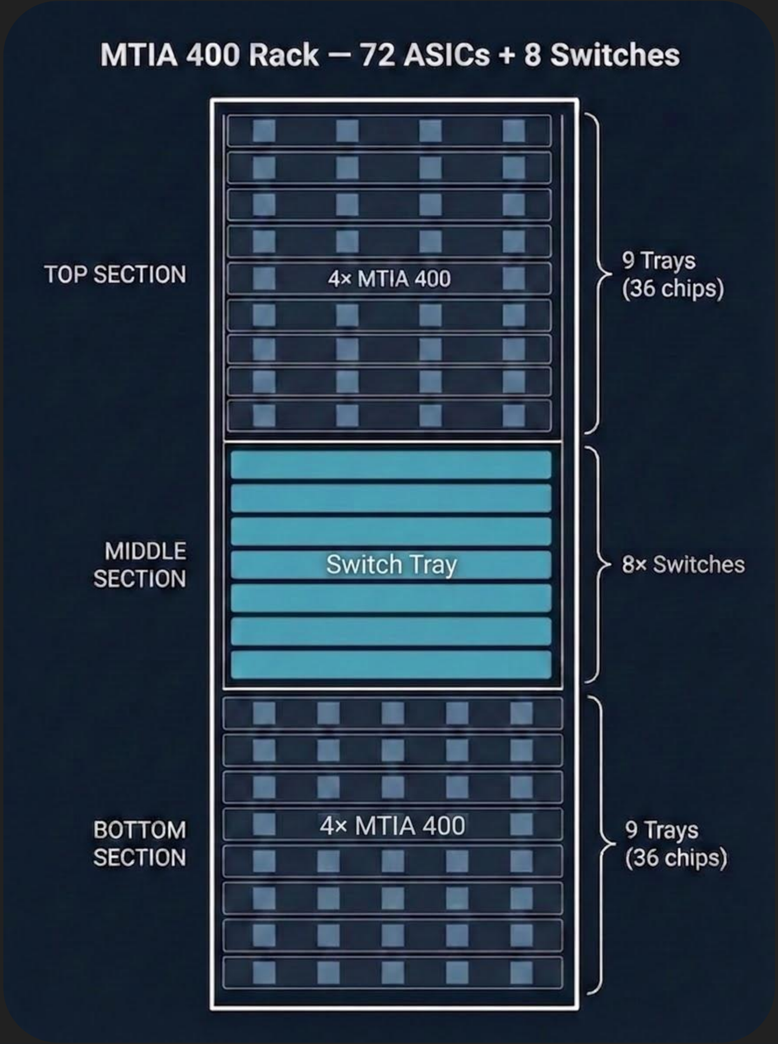
- Native read-reduce-write in LLC SRAM
- Supported multiple data formats
- Operations: min, max, sum
- Eliminates data movement overhead for collecti op



System Design

- 4x MTIA 400 per compute tray
- 72 ASICs in Scale-up domain





MTIA 400 Performance

3 PFLOPS FP16/BF16

6 PFLOPS FP8/MX8

12 PFLOPS MX4

vs MTIA 300

~5x FP16 compute (0.6 to 3 PFLOPS)

1.3x HBM capacity (216 to 288 GB)

1.5x HBM bandwidth (6.1 to 9.4 TB/s)

1.0x I/O bandwidth (1.2 TB/s)

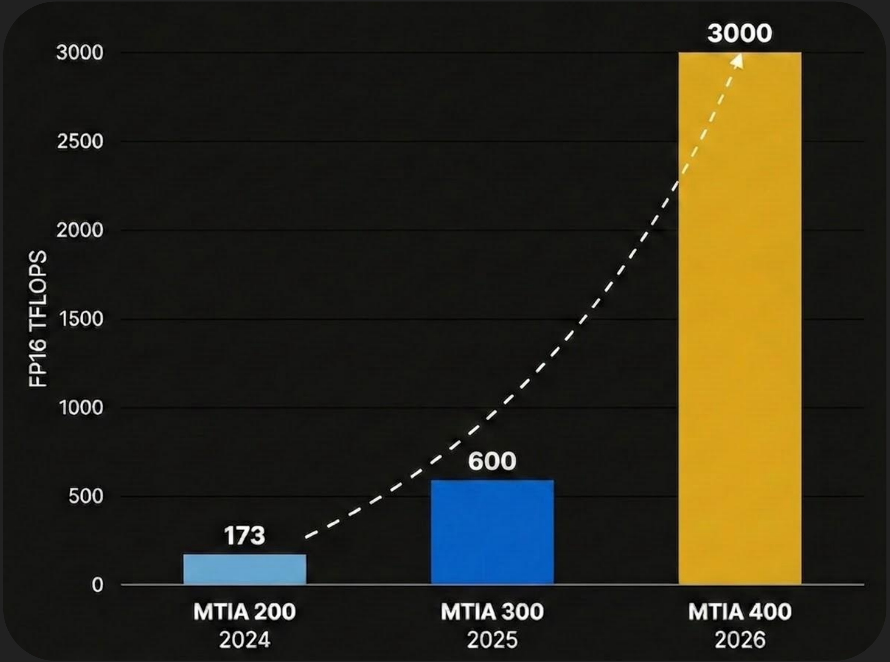
vs MTIA 200

15.4x FP16 compute

46x DRAM bandwidth (HBM3E)

16x host PCIe bandwidth

5x on-chip SRAM bandwidth



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Conclusions and What's Next

MTIA 400 for both DLRM and GenAI

5-chiplet 2.5D package — scales beyond monolithic limits

4x MAC density PE grid with native MX format, horizontal reduction

HEC and PEC for DLRM

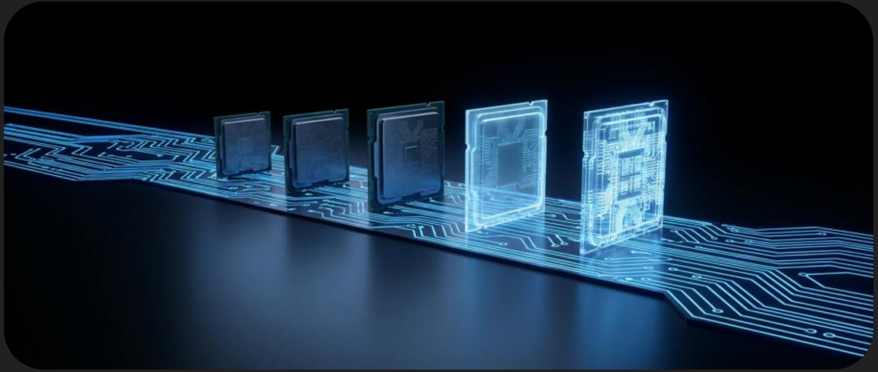
Dedicated ME with NMC, SGM, and LLC reduction for collectives

1.2 TB/s Scale-up BW

What's next (see MTIA roadmap)

MTIA 450 — Enhanced GenAI inference

MTIA 500 — Further GenAI inference scaling



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Q & A

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